

MATH 124 Tutor Guide

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This is a tutor guide for MATH 124 at UW. It contains hints and comments to selected problems in the homework and worksheets. It is intended to give the tutors a quick head-up on the most-often-asked questions.

The structure may be weird since I start with Week 2. This is due to the first week of Fall quarter 2020 began on a Friday.

Here I might add a content page..

1 Week 2

Material covered. tangents to circles and other curves; parametric equations;

cf. Section 10.1, 2.1

Problems. Here are some types of problem which are often found at MSC.

- **Finding tangent lines to a circle through a given point without calculus**

Let the given point be (s, t) and the circle be $(x - c)^2 + (y - d)^2 = r^2$. Assume the tangent point being (a, b) . Since (a, b) lies on the circle, it satisfies the circle equation. Set up another equation using the fact the tangent line is perpendicular to the radius. The product of their slopes is -1 , and this gives us a second equation. With two equations in a and b , we can solve for a and b .

- **Finding tangent line(s) at a given point of a given parabola without calculus**

Let $f(x) = ax^2 + bx + c$ and the given point be $(z, f(z))$. Write the tangent line L as $y = m(x - z) + f(z)$. Then all intersection points of L and the curve f can be found by solving the equation $f(x) = m(x - z) + f(z)$, which is a quadratic equation in x . Because L being the tangent line only intersects the graph of f at exactly one point, set the discriminant equaling 0 and solve for m .

- **Finding when two particles collide using parametric equations**

On a plane, suppose we have two particles (x_1, y_1) and (x_2, y_2) whose locations are given by the following parametric equations

$$\begin{aligned}x_1 &= a_1 \sin t + b_1 \cos t, & y_1 &= c_1 \sin t + d_1 \cos t, \\x_2 &= a_2 \sin t + b_2 \cos t, & y_2 &= c_2 \sin t + d_2 \cos t.\end{aligned}$$

Equate the x, y coordinates and substitute $\sin t = u$ and $\cos t = v$ in the system below

$$\begin{aligned}(a_1 - a_2) \sin t + (b_1 - b_2) \cos t &= 0, \\(c_1 - c_2) \sin t + (d_1 - d_2) \cos t &= 0.\end{aligned}$$

Solve for u, v and then find t such that $\sin t = u$ and $\cos t = v$ hold simultaneously.

2 Week 3

Material covered. limits; continuity; asymptotes;

cf. Section 2.2-2.3, 2.5-2.6

Problems. Here are some types of problem which are often found at MSC.

- **Find a limit of a fraction**

- Start with taking the limit of the numerator and the limit of the denominator. If both evaluate to a number, then we can simply read off the limit.
- If the fraction is of the forms such as $\frac{a}{\pm\infty}$ (or $\frac{\pm\infty}{a}$), then the limit is zero (or $\pm\infty$.)
- But if the fraction is $\frac{a}{0}$, we can't conclude anything.

$$\text{Eg. } \lim_{t \rightarrow \pi/2} \frac{\sin t + \sqrt{\sin^2 t + 4 \cos^2 t}}{2 \cos^2 t} = \frac{2}{0^+} \rightarrow \infty.$$

$$\text{Eg. } \lim_{t \rightarrow \pi/2} \frac{\sin t + \sqrt{\sin^2 t + 4 \cos^2 t}}{2 \cos t} \text{ does not exist because:}$$

$$\lim_{t \rightarrow \pi/2^+} \frac{\sin t + \sqrt{\sin^2 t + 4 \cos^2 t}}{2 \cos t} = \frac{2}{0^-} = -\infty \text{ but}$$

$$\lim_{t \rightarrow \pi/2^-} \frac{\sin t + \sqrt{\sin^2 t + 4 \cos^2 t}}{2 \cos t} = \frac{2}{0^+} = +\infty.$$

- **Finding a limit in an indeterminate form such as $\frac{0}{0}$ or $\frac{\pm\infty}{\pm\infty}$.**

- If the denominator and the numerators are both polynomials, we can factor those polynomials and cancel all the terms we can.

$$\text{Eg. } \lim_{x \rightarrow 5} \frac{x^2 - 9x + 20}{x - 5} = \lim_{x \rightarrow 5} \frac{(x - 5)(x - 4)}{x - 5} = \lim_{x \rightarrow 5} (x - 4) = -1.$$

- If there is a radical expression in either the denominator or the numerator, use rationalization. If multiple radical expressions appear in both the denominator and the numerator, use double rationalization, i.e., rationalization with respect to all radical expressions.

$$\text{Eg. } \lim_{x \rightarrow 8} \frac{\sqrt{12 - x} - 2}{\sqrt{24 - x} - 4} = \lim_{x \rightarrow 8} \frac{\sqrt{12 - x} - 2}{\sqrt{24 - x} - 4} \cdot \frac{\sqrt{12 - x} + 2}{\sqrt{12 - x} + 2} \cdot \frac{\sqrt{24 - x} + 4}{\sqrt{24 - x} + 4} = \lim_{x \rightarrow 8} \frac{\sqrt{24 - x} + 4}{\sqrt{12 - x} + 2}$$

$$\text{Eg. } \lim_{t \rightarrow \pi/2} \frac{\sin t - \sqrt{\sin^2 t + 4 \cos^2 t}}{2 \cos^2 t} \text{ is an indeterminate form } \frac{0}{0}. \text{ So use rationalization.}$$

- **Finding the limit of a fraction at infinity**

When the numerator and the denominator are polynomials (or linear combination of rational powers), we can evaluate the limit after dividing by the highest power of the variable.

$$\text{Eg. } \lim_{t \rightarrow \infty} \frac{\sqrt{t} + t^2}{5t - t^2} = \lim_{t \rightarrow \infty} \frac{t^{-3/2} + 1}{5t^{-1} - 1} = \lim_{t \rightarrow \infty} \frac{1}{-1} = -1.$$

3 Week 4

Material covered. derivatives, derivative functions;

cf. Section 2.7-2.8

Problems. Here are some types of problem which are often found at MSC.

- **Differentiating a quotient of polynomials using the definition of derivation**

One algebraic trick that might ease the tedious calculation is to use polynomial division and separate out a constant.

Eg. Note $\frac{7+x}{1-7x} = \frac{(-\frac{1}{7} + x) + \frac{50}{7}}{1-7x} = \frac{1}{7} + \frac{\frac{50}{7}}{1-7x}$. Now $(\frac{7+x}{1-7x})' = \lim_{h \rightarrow 0} \frac{50/7}{1-7(x+h)} - \frac{50/7}{1-7x}$.

Compare our result with $(\frac{7+x}{1-7x})' = \lim_{h \rightarrow 0} \frac{\frac{7+(x+h)}{1-7(x+h)} - \frac{7+x}{1-7x}}{h}$ which is tedious.

- **Finding the derivative of an ellipse without implicit differentiation**

Suppose $ax^2 + by^2 = c$. The equation above is a quadratic equation in y . So we solve $y = \pm \sqrt{\frac{c-ax^2}{b}}$. Choose the correct piece based on in which quadrant it lies. We compute the

derivative as $\lim_{h \rightarrow 0} \frac{\pm \sqrt{\frac{c-a(x+h)^2}{b}} \mp \sqrt{\frac{c-ax^2}{b}}}{h}$. Proceed by rationalizing the numerator.

- **Finding informations of a location given by parametric equations**

Suppose the location of an object is given by linear parametric equations $x(t) = a + m_1 t$ and $y(t) = b + m_2 t$. The derivatives of $x'(t) = m_1$ and $y'(t) = m_2$ give the horizontal and vertical instantaneous rate of change (velocity). The average velocity equals the instantaneous velocity. Note we can solve $t = (x - a)/m_1$ and plug it in $y(t)$ to get $y = \frac{m_2}{m_1}(x - a) + b$. The velocity of this object is $\sqrt{x'^2 + y'^2} = \sqrt{m_1^2 + m_2^2}$ (which is a constant and also implies this object moves along a line).

4 Week 5

Material covered. derivative rules; trig derivatives;

cf. Section 3.1-3.3

Problems. Here are some types of problem which are often found at MSC.

- **Finding tangent line(s) of a given curve passing through a given point**

Let the curve be $f(x)$ and the given point be (s, t) . Assume the tangent point be (a, b) . Then $f'(a) = \frac{t-b}{s-a}$. Since $b = f(a)$, we have two equations in two unknowns a, b , which we can solve for a and b . The tangent line L is $y = f'(a)(x - a) + b$.

- The question often specifies the x or y intercept of L . These conditions are equivalent to L passing through $(s, 0)$ or $(0, t)$.

- **Finding tangent line(s) of a given curve parallel to a given line**

Suppose the given line has slope m . Assume the tangent point be (a, b) . Solve $f'(a) = m$ for a to find the tangent points on the curve, at which the tangent line is parallel.

5 Week 6

Material covered. chain rule; derivatives for parametric equations;

cf. Section 3.4, Section 10.2

Problems. Here are some types of problem which are often found at MSC.

- **Finding line(s) tangent to two curves simultaneously**

Let f, g be two curves. Suppose the line L which intersects the curves at $(a, f(a))$ and $(b, g(b))$. Because the slope of L equals the derivative at each point, we can set up $f'(a) = g'(b) = \frac{g(b) - f(a)}{b - a}$. Choose two equations from the above equality and solve for a and b .

- **Finding tangent line(s) to a parametrized curve passing through a given point**

Suppose the point of tangency is $(f(a), g(a))$ when $t = a$, and the given point is (u, v) . Then solve $\frac{g'(a)}{f'(a)} = \frac{g(a) - v}{f(a) - u}$ for a .

- **Finding tangent line(s) to an ellipse passing through a given point without implicit differentiation**

Suppose the ellipse is given by $u^2x^2 + v^2y^2 = w^2$. We convert the ellipse to the following parametric equations

$$x(t) = \frac{\cos t}{u} \quad y(t) = \frac{\sin t}{v},$$

and we go back to bulletin 2.

- **Finding informations of a particle whose location is given by parametric equations**

Suppose the particle's location is described as $x(t)$ and $y(t)$. Then $x'(t)$ and $y'(t)$ are the horizontal and vertical velocity. The speed of the particle is $\sqrt{x'(t)^2 + y'(t)^2}$. The magnitude of the angle at which the particle hits the ground is $\tan^{-1} \frac{y'(t)}{x'(t)}$. If the particle passes through a point (s, t) , then we use $x(t) = s$ to solve for t . Plug in t we get all the quantities above.

6 Week 7

Material covered. implicit differentiation; logarithmic differentiation; related rates

cf. Section 3.5-3.9

Problems. Here are some types of problem which are often found at MSC.

- **Finding information of an implicit function**

Suppose we are given an implicit function $F(x, y) = 0$

- To find its x or y-intercept, we set $y = 0$ (or $x = 0$) and solve $F(0, y) = 0$ (or $F(x, 0) = 0$).
- If point $P = (s, t)$ lies in $F(x, y) = 0$, then the tangent line passing through P is $y = \frac{dy}{dx}|_{x=s}(x - s) + b$. Use implicit differentiation to find $\frac{dy}{dx}|_{x=s}$.
- To find the tangent line of $F(x, y) = 0$ passes through a given point $Q = (s, t)$, assume the tangent point is (a, b) . We then have two equations $t = \frac{dy}{dx}|_{x=a}(a - s) + b$ and $F(a, b) = 0$ in two unknowns. Solve for a, b .

- **Finding information of a particle whose location is given by parametric equations**

Suppose the particle's location is described as $x(t)$ and $y(t)$.

- Then $x'(t)$ and $y'(t)$ are the horizontal and vertical velocity.
- The speed of the particle is $\sqrt{x'(t)^2 + y'(t)^2}$.
- If point $P = (a, b)$ lies in $F(x, y) = 0$, then the slope of the tangent line passing through P is $\frac{dy}{dx}|_{x=a}$. Use implicit differentiation to find $\frac{dy}{dx}|_{x=s}$.
- The tangent line through P is $y = \frac{dy}{dx}|_{x=a}(x - a) + b$.
- To find when the tangent line is horizontal, we set $\frac{dy}{dx}|_{x=a} = \frac{y'(a)}{x'(a)} = 0$ and solve for t . This is equivalent to solving $y'(a) = 0$ and $x'(a) \neq 0$.
- To find when the tangent line is vertical, we set $\frac{dy}{dx}|_{x=a} = \frac{y'(a)}{x'(a)} = \infty$ and solve for t . This is equivalent to solving $y'(a) \neq 0$ and $x'(a) = 0$. One need to be careful with the case when $y'(a) = 0$ and $x'(a) = 0$ in order to see what is the slope of the tangent line. This is a case where the limit has $\frac{0}{0}$ indeterminate form.

7 Week 8

Material covered. related rates; linear approximation, extreme values of a function

cf. Section 3.9-10, Section 4.1, law of cosine

Problems. Here are some types of problem which are often found at MSC.

- **Rate of change of the height of a firefighter on a ladder**

9. Question Details

firefighter [1640590]

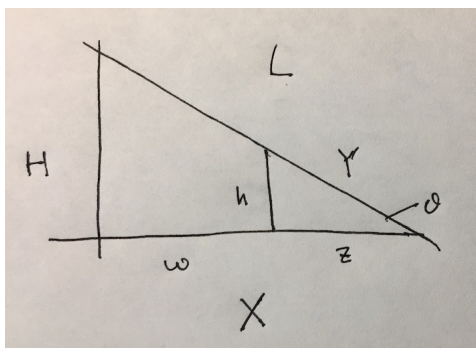
A ladder 28 feet long leans against a wall and the foot of the ladder is sliding away at a constant rate of 3 feet/sec. Meanwhile, a firefighter is climbing up the ladder at a rate of 2 feet/sec. When the firefighter has climbed up 6 feet of the ladder, the ladder makes an angle of $\pi/3$ with the ground. Answer the two related rates questions below. (Hint: Use two carefully labeled similar right triangles.)

(a) If h is the height of the firefighter above the ground, at the instant the angle of the ladder with the ground is $\pi/3$, find $dh/dt =$ feet/sec.

(b) If w is the horizontal distance from the firefighter to the wall, at the instant the angle of the ladder with the ground is $\pi/3$, find $dw/dt =$ feet/sec.

Let L be the length of the ladder. Let $X(t)$ be the distance from the foot of the ladder to the wall. Let $H(t)$ be the height of the ladder. Let $Y(t)$ be the distance travelled by the firefighter along the ladder. Let $h(t), w(t), \theta(t)$ be defined in the figure above. Let $z(t)$ be the horizontal distance from the firefighter to the foot of the ladder. Suppose $Y, \theta, \frac{dY}{dt}$ and $\frac{dX}{dt}$ are given.

– We want to solve $\frac{dh}{dt}$ in terms of $\frac{dY}{dt}$ and $\frac{dX}{dt}$. Draw the graph below.



– By pythagorean theorem $Y^2 = h^2 + z^2$ which implies $2Y \frac{dY}{dt} = 2h \frac{dh}{dt} + 2z \frac{dz}{dt} (*)$.

– Using similarity of triangles, we get $\frac{z}{X} = \frac{Y}{L}$ which implies $z = \frac{1}{L}XY$.

– By product rule that $\frac{dz}{dt} = \frac{1}{L} \left(\frac{dX}{dt}Y + \frac{dY}{dt}X \right)$.

– Plug $\frac{dz}{dt}$ in $(*)$, and we solve for $\frac{dh}{dt}$.

– Note $w = X - z$. So it follows $\frac{dw}{dt} = \frac{dX}{dt} - \frac{dz}{dt}$.

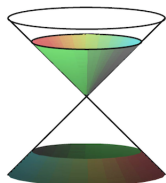
• Rate of change of the height of the sand

10. Question Details

hourglassrates [2520960]

(Old Final Exam Problem-Sp2009)

An hourglass is made up of two glass cones connected at their tips. Both cones have radius 1 inch and height 3 inches. When the hourglass is flipped over, sand starts falling to the lower cone.



(a) When the sand remaining in the **upper cone** has height y inches, its volume A in terms of y is .

(b) When the sand in the **lower cone** has reached a height of h inches, its volume B in terms of h is .
(Hint: B is the volume of the bottom cone minus the volume of the empty space above the sand.)

(c) Assume the total volume of sand in the hourglass is $3\pi/4$ cubic inches. Also, assume the height of the sand in the upper cone is decreasing at a rate of $15/100$ inches per second. At the instant when the sand in the lower cone is 1 inch high, the height of the sand in the lower cone is increasing at a rate of inches per second.

An hourglass has the form of two cones meeting at their vertices, each with radius R and height H . Suppose the total volume of sand in the hourglass is V_0 . Let y be the height of sand in the upper cone and h be the height of sand in the lower cone. Suppose h and $\frac{dy}{dt}$ are given. We want to find $\frac{dh}{dt}$.

Using similar triangles, the volume of sand in the upper cone is

$$V_1 = \frac{1}{3}\pi\left(\frac{R}{H}y\right)^2y = \frac{1}{3}\pi\frac{R^2}{H^2}y^3.$$

The volume of sand in the lower cone is

$$V_2 = \frac{1}{3}\pi R^2 H - \frac{1}{3}\pi\left(\frac{R}{H}(H-h)\right)^2(H-h) = \frac{1}{3}\pi R^2 H - \frac{1}{3}\pi\frac{R^2}{H}(H-h)^3.$$

We have $V_0 = V_1 + V_2$ which we can differentiate and simplify to

$$0 = y^2\frac{dy}{dt} + (H-h)^2\frac{dh}{dt}.$$

Plug in h and $\frac{dy}{dt}$ and solve for $\frac{dh}{dt}$.

• Linear approximation

Suppose two variables x, y are related by $F(x, y) = 0$. Treat y be a function of x . Use implicit differentiation to find $\frac{dy}{dx}$. Suppose (x_0, y_0) is a point on $F(x, y) = 0$.

- Find $m = \frac{dy}{dx}$ evaluated at $P = (x_0, y_0)$.
- The linear approximation of y at P is $y = m(x - x_0) + y_0$.
- if x is given, plug in x and compute y ;
- if y is given, we can solve for $x = \frac{y - y_0 + mx_0}{m}$ and solve for $\Delta x = x - x_0 = \frac{y - y_0}{m}$.

8 Week 9

Thanksgiving break

9 Week 10

Material covered. extreme values; derivatives and curve tracing; L'Hopital's rule

cf. Section 4.1-4.4

Problems. Here are some types of problem which are often found at MSC.

- **Extreme values of a continuous function**

To find the extreme values of $f(x)$, we go through the following steps (pp. 281 Stewart)

- find $f'(x)$;
- find all solutions to $f'(x) = 0$ and where $f'(x)$ does not exist, i.e., the critical numbers;
- check $f(x)$ at the endpoints;

E.g. For $F(x) = x^{4/5}(x-4)^2$, its derivative $F'(x) = \frac{4}{5}x^{-1/5}(x-4)^2 + x^{4/5}2(x-4)$ is not defined at 0. So 0 is a critical number since F is clearly defined on entire $\mathbb{R}$.

E.g. For $f(x) = x^{-6}\ln(x)$, its derivative $-6x^{-7}\ln(x) + x^{-6}\frac{1}{x}$ is not defined on 0. But f is not defined on 0 too because of $\ln(x)$. So 0 is not a critical number.

- **Undetermined coefficient of a function**

Suppose a function $f(x)$ attains local extreme values y_i at x_i . Assume there are k unknown coefficients in f and we want to solve for them.

- Set up equations $f'(x_i) = 0$ and $f(x_i) = y_i$.
- There are $2i$ equations in k unknowns. If $k \leq 2i$, the system is solvable.

E.g. For a polynomial $f(x) = \sum_{i=0}^n ax^3 + bx^2 + cx^1 + d$ attains local extreme values y_i at x_i for $i = 1, 2$. We can solve the 4 equations $f'(x_i) = 0$ and $f(x_i) = y_i$ in 4 unknowns a, b, c, d .

- **y-intercept of the normal lines**

Let P be a point on the graph of $y = e^{-x^2/k}$ for some $k > 0$ with nonzero x -intercept a . Suppose b is the y -intercept of the normal line through P . We want to find $\lim_{a \rightarrow 0} b$ and $\lim_{a \rightarrow \infty} b$.

- The normal line is perpendicular to the tangent line. So it is $y = \frac{k}{2a}e^{a^2/k}(x-a) + e^{-a^2/k}$.
- Plug in $x = 0$, we get $b = -\frac{k}{2}e^{a^2/k} + e^{-a^2/k}$. So $\lim_{a \rightarrow 0} b = \lim_{a \rightarrow 0} -\frac{k}{2}e^{a^2/k} + e^{-a^2/k} = -\frac{k}{2} + 1$.
- Meanwhile $\lim_{a \rightarrow \infty} b = \lim_{a \rightarrow 0} -\frac{k}{2}e^{a^2/k} + e^{-a^2/k} = -\infty$.

- **Logarithm differentiation**

There is a technique to compute the limit of a composed function.

E.g. Find $\lim_{x \rightarrow 0} (1+3x)^{3/x}$.

Let $y = (1+3x)^{3/x}$. Then $\log y = \frac{3}{x} \log(1+3x)$. It follows

$$\lim_{x \rightarrow 0} \log y = \lim_{x \rightarrow 0} \frac{3 \log(1+3x)}{x} \stackrel{H}{=} \lim_{x \rightarrow 0} \frac{3 \frac{1}{1+3x}}{1} = 3.$$

The last step is due to L'Hopital's rule. Because exp is continuous,

$$\lim_{x \rightarrow 0} y = \lim_{x \rightarrow 0} e^{\log y} = e^{\lim_{x \rightarrow 0} \log y} = e^3.$$

10 Week 11

Material covered. extreme values; derivatives and curve sketching; L'Hopital's rule

cf. Section 4.1-4.4

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- find $f'(x)$;
- find all solutions to $f'(x) = 0$ and where $f'(x)$ does not exist, i.e., the critical numbers;
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- Set up equations $f'(x_i) = 0$ and $f(x_i) = y_i$.
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E.g. For a polynomial $f(x) = \sum_{i=0}^n ax^3 + bx^2 + cx^1 + d$ attains local extreme values y_i at x_i for $i = 1, 2$. We can solve the 4 equations $f'(x_i) = 0$ and $f(x_i) = y_i$ in 4 unknowns a, b, c, d .

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- The normal line is perpendicular to the tangent line. So it is $y = \frac{k}{2a}e^{a^2/k}(x-a) + e^{-a^2/k}$.
- Plug in $x = 0$, we get $b = -\frac{k}{2}e^{a^2/k} + e^{-a^2/k}$. So $\lim_{a \rightarrow 0} b = \lim_{a \rightarrow 0} -\frac{k}{2}e^{a^2/k} + e^{-a^2/k} = -\frac{k}{2} + 1$.
- Meanwhile $\lim_{a \rightarrow \infty} b = \lim_{a \rightarrow 0} -\frac{k}{2}e^{a^2/k} + e^{-a^2/k} = -\infty$.

- **Logarithm differentiation**

There is a technique to compute the limit of a composed function.

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The last step is due to L'Hopital's rule. Because exp is continuous,

$$\lim_{x \rightarrow 0} y = \lim_{x \rightarrow 0} e^{\log y} = e^{\lim_{x \rightarrow 0} \log y} = e^3.$$

11 Week 12

Material covered. derivatives and curve sketching; optimization; some past final exams;

cf. Section 4.4-4.5, 4.7, some past final exams

Problems. Here are some types of problem which are often found at MSC.

- **Undetermined coefficient of a parametric equation**

The motion of a point on a bicycle wheel is given by the parametric equation,

$$x(t) = At + \cos 2t \quad y(t) = B - \sin 2t$$

where A, B are positive numbers. The y -coordinate of the point is zero when it is at the bottom, and there the horizontal velocity is zero. Find A, B .

- At each fixed t , the point $(x(t), y(t)) = (At, B) + (\cos 2t, -\sin 2t)$ is a point at the circle of radius 1 centered at (At, B) . Then lowest point has height $B - 1$, which is zero iff $B = 0$.
- To solve A , we use $x'(t) = A - 2\sin 2t = 0$ and the fact $\sin 2t = 1$ when the point is at the bottom.

- **Tangent line of a parametric equation**

A curve is given by parametric equations $x(t) = f(t)$ and $y(t) = g(t)$. Find its tangent line passing through a given point c, d .

- Let the tangent point be $(x(t), y(t))$ and the slope of the tangent line be m . We note $m = \frac{y(t)-d}{x(t)-c}$.
- Recall $m = \frac{dy}{dx}|_{(x(t), y(t))} = \frac{dy/dt}{dx/dt}|_t$ by chain rule and solve for t by $\frac{dy/dt}{dx/dt}|_t = \frac{y(t)-d}{x(t)-c}$.

- **Computing limit**

Compute $\lim_{x \rightarrow 0^+} \frac{12}{x} - \frac{12}{x^2} + \frac{4}{x^3}$.

Because of limit law and continuity of $\frac{12}{x} - \frac{12}{x^2} + \frac{4}{x^3}$ on $(0, \infty)$, we get

$$\begin{aligned} \lim_{x \rightarrow 0^+} \frac{12}{x} - \frac{12}{x^2} + \frac{4}{x^3} &= \lim_{x \rightarrow 0^+} \frac{1}{x^3} (12x^2 - 12x + 4) \\ &= \left(\lim_{x \rightarrow 0^+} \frac{1}{x^3} \right) \left(\lim_{x \rightarrow 0^+} 12x^2 - 12x + 4 \right) = +\infty \cdot 4 = \infty. \end{aligned}$$

- **Growth rate of $\ln(x)$, x^n , and c^x**

Compute $\lim_{x \rightarrow \infty} \frac{5x^2 3^{1+x} + 3x^4 \ln(x)}{2x^2 3^{3+x} + 9x^4 \ln(x)}$.

- Use L'Hopital's rule to show $\lim_{x \rightarrow \infty} \frac{\ln(x)}{x^n} = 0$ and $\lim_{x \rightarrow \infty} \frac{x^n}{c^x} = 0$.
- Divide through the term grows fastest, which is $2x^2 3^{3+x}$ and get

$$\lim_{x \rightarrow \infty} \frac{(5x^2 3^{1+x})/(2x^2 3^{3+x}) + (3x^4 \ln(x))/(2x^2 3^{3+x})}{1 + (9x^4 \ln(x))/(2x^2 3^{3+x})} = 0.$$